

PROJECT REPORT: HOSPITAL AGENTIC AI

TASK 2: Fine-Tuned Transformer for Medical Classification

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1 OBJECTIVE

The objective of Task-2 is to enhance the Hospital AI Assistant by integrating a **Fine-Tuned Transformer Model**. This model automatically classifies patient symptom queries into appropriate medical departments, replacing simple keyword matching with semantic understanding.

2 DATASET DESCRIPTION

A large healthcare dataset containing free-text symptom descriptions and disease labels was processed and mapped into **10 Medical Departments**.

Target Classes (Departments):

- Cardiology
- Dermatology
- ENT (Ear, Nose, Throat)
- Endocrinology
- Gastroenterology
- General Medicine
- Neurology
- Orthopedics
- Psychiatry
- Pulmonology

Dataset Statistics:	
Metric	Value
Total Samples	~ 23,700
Training Split	80%
Validation Split	20%
Input Type	Free-text

3 MODEL SELECTION & TRAINING SETUP

Model Architecture

We selected **DistilBERT (distilbert-base-uncased)** for the following reasons:

- **Efficiency:** Significantly lighter and faster than full BERT.
- **Resource Constraint:** Suitable for training on limited GPU resources (Google Colab).
- **Performance:** Proven capability in text classification tasks.

Hyperparameters

Parameter	Value	Parameter	Value
Framework	Hugging Face	Optimizer	AdamW
Epochs	3	Batch Size	16
Loss Function	Cross-Entropy	Device	GPU (T4)

Training Results

- **Loss Convergence:** Training loss decreased gradually, while validation loss remained stable.
- **Overfitting:** No significant overfitting was observed due to the low epoch count.

4 CONFUSION MATRIX ANALYSIS

A confusion matrix was generated to evaluate class-wise performance. The results highlight the complexity of medical diagnosis from short text.

Observations

Strong Performance	ENT, Dermatology, Orthopedics. These departments have distinctive keywords (e.g., "skin rash", "joint pain") making them easier to classify.
Moderate Performance	Gastroenterology, Pulmonology. Showed reasonable detection but some overlap with General Medicine.
Weak Performance	Cardiology, General Medicine. Frequent misclassification due to generic symptoms (e.g., "fatigue", "pain") that appear across multiple specialties.

Quantitative Metrics

Metric	Value	Interpretation
Accuracy	47%	Acceptable for a 10-class prototype with noisy data.
Macro F1-Score	0.30	Impacted by underperforming minority classes.
Weighted F1-Score	0.36	Higher due to good performance on dominant classes.

Key Insight

"The confusion matrix highlights the inherent ambiguity of medical symptoms. While specific categories like ENT are predicted accurately, overlap among general symptoms leads to misclassification in broader departments like General Medicine."

5 INFERENCE & INTEGRATION

The trained model is integrated into the Task-1 pipeline to replace the rule-based intent classifier.

Inference Example

```
> Input Query: "I have severe chest pain and breathlessness"  
> Tokenizer Output: [101, 1045, 2031, ... 102]  
> Model Prediction: Cardiology  
  
> System Action: Routing user to Dr. Sharma (Cardiologist)...
```

6 CHALLENGES & CONCLUSION

Challenges Faced

- **Class Imbalance:** Some diseases had significantly fewer samples.
- **Symptom Overlap:** "Fever" or "Pain" can belong to almost any department.
- **Resource Limits:** Training was restricted to 3 epochs due to hardware constraints.

Conclusion

Task-2 successfully demonstrates the use of a **Fine-Tuned Transformer** for medical text classification. While accuracy is limited by data constraints, the model provides a scalable, intelligent foundation for routing patients, significantly improving upon the rule-based approach of Task-1.